

INSTITUTION: Sir Padampat Singhania University (SPSU), Udaipur	**DOCUMENT REF:** SPSU/CDC/DIR/2026/ADM-01
UNIT: Career Development Cell (CDC) — Directorate Office	**SECURITY:** STRICTLY CONFIDENTIAL • INTERNAL
SUBJECT: Official Portal Administrator & Operations Manual	**EFFECTIVE DATE:** October 2026 (Annual Release)

SPSU Career Development Cell (CDC)

Power BI Executive Intelligence Portal — Administrator Guide

1. Executive Summary & Architecture

The **SPSU Career Development Cell (CDC) Power BI Executive Intelligence Portal** is a state-of-the-art decision-support and placement management platform engineered specifically for the **Director, Vice Chancellor, and Career Development Cell leadership**. It provides real-time transparency into graduating batch placement conversions, summer internship commitments, compensation distributions, and corporate recruiter partnerships.

Core Architecture & Design Language: Built on a modern React 19 and Vite 6 foundation styled with Microsoft Power BI executive design patterns (fluent header ribbon, dynamic filter slicers, KPI performance cards, drill-through student profile drawers, and instant multi-field query engine).

Component	Technology / Specification	Operational Purpose
Frontend Engine	React 19 + Vite 6 + Tailwind CSS	Sub-second client rendering & fluid transitions
Design System	Power BI Executive Edition	Directorate presentation standards & interactive slicers
Dataset Size	392 Students / 9 Academic Disciplines	CSE (192), ECE (18), MN (17), CE (11), ME (10), BDES (2), BCA (38),
Hosting Architecture	GitHub Pages Cloud + Offline Standalone	Dual live cloud URL + zero-dependency standalone HTML
Security Protocol	Role-Gated Authentication Gate	Directorate access with session token & logout control

2. Access Control & Administrator Credentials

Access to the executive intelligence console is strictly restricted through the **SPSU Authentication Gate**. The portal validates credentials against the institutional administrative registry and issues session tokens.

AUTHENTICATION GATE CREDENTIALS	ACCESS LEVEL: Directorate Executive / Super Admin
Username: `ADMIN` (Case-insensitive)	**Password: `SPSU123`**
Auto-Fill Shortcut: Click *'Auto-fill Credentials'* for instant 1-click administrative authorization.	**Session Persistence:** Optional *'Keep me logged in'* check saves secure browser session token.

Sign-Out & Session Revocation:

- **Direct Sign Out:** Click the red **Log Out icon** located in the top-right header ribbon to immediately revoke authentication tokens and return to the secure login gate.
- **Role Switcher:** Click the User Profile badge ('■■■ *Administrator / Super Admin*') to open the Auth Gateway modal to preview department auditor or recruiter views.

3. Portal Modules & Intelligence Directory

The portal is organized into seven specialized executive views accessible via the primary ribbon navigation:

Ribbon Tab	Core Intelligence Capabilities	Primary Target Audience
Executive Cockpit	High-level KPI scorecard (Placements, SI, Peak CTC ■10.2 LPA, Avg CTC ■5.85 LPA), top recruiters & accreditations showcase,	Director & Vice Chancellor
Placement Hub	FTE vs PPO breakdown, Tier 1 (>=8 LPA), Tier 2 (5-8 LPA), Tier 3 (<5 LPA) package tiers, and placement offers, and company offer	CDC Head & Placement Officers
Summer Internship	Summer internship conversion rates (75.8%), cumulative 374 industry offers, stipend tracking, and male vs female cohort parity metrics.	Internship Coordinators
Student 360° Matrix	Granular student grid with multi-field search, numerical sorting (CTC, CGPA, Backlogs, Status), and individual drill-dow	CDC Staff & Academic Deans
Recruiter Intelligence	Tier 1 hiring partners (KMK Venture, Country Edu, Xebia, Celebal, Eleation), offer distribution, Relationship Manager, and corporate drive history.	Corporate Relations Managers
Reports Studio	Official Director's 1-Pager Executive Memo generator, CSV data exports, Department-wise analytics, and ready PDF memo formatting.	Executive Assistants & Deans
Template Manager	Annual rollover data ingestion engine, drag-and-drop CSV master upload, and schema validation	Database & IT Administrators

4. Interactive Power BI Slicers & Query Operations

A cornerstone of the executive portal is the **Collapsible Power BI Slicers Pane**, accessible via the **'Slicers'** button in the top ribbon.

- **Discipline / Program Slicer:** Filter all dashboard views across 9 degree disciplines (CSE, ECE, MN, CE, ME, BDES, BCA, BBA, MBA) or select 'ALL' for institutional aggregate.
- **Placement Status Slicer:** Segment cohorts into Placed-FTE, Placed-PPO, Remaining Unplaced, or Opt-Out categories.
- **Summer Internship Slicer:** Isolate students who have secured internships vs. those awaiting placement.
- **Instant Global Slicer Reset:** Click the **'Reset'** button in the ribbon to restore default parameters across all 392 students.

Student 360° Matrix — Deep Search & Numerical Sorting:

The Student 360° Matrix features an upgraded search and sorting engine designed to handle large institutional datasets:

1. **Multi-Field Instant Search:** Type any keyword in the search bar. The engine checks Student Name, Enrollment Number (e.g. 2223010001), SPSU Email, Department, Specialization (Cloud, AI/ML, Cyber Security), and Recruiter Companies simultaneously.
2. **True Numerical Sorting:** Click column headers (**S.No, CTC, CGPA, Backlogs**) to sort in accurate numerical sequence (e.g., 10.20 LPA → 8.00 LPA → 6.50 LPA) rather than alphabetical text sorting.
3. **Automated Pagination Reset:** Whenever a search term or department filter is applied, the grid automatically resets to Page 1 to prevent empty-page display anomalies.
4. **Drill-Down Profile Drawer:** Click the **'View Profile'** eye icon next to any student to open their comprehensive 360° Profile Dossier containing verified backlogs, internship offers, company CTC, and contact coordinates.

5. Export Engine & 1-Click Executive Memo Generation

The Directorate portal incorporates an automated multi-format export pipeline to facilitate executive reporting:

Export Option	Output Format	Description & Intended Usage
Director's 1-Pager Memo	**PDF / Print Dialog**	Generates an official executive briefing document with embedded university banner, key KPI metrics, program-wise placement table, and formal signature lines for the Director.
Filtered Students Dataset	**CSV Spreadsheet**	Downloads all current filtered student records with full attributes (Enrollment, CGPA, CTC, Company, Backlogs) formatted for Microsoft Excel or Google Sheets.
Program Summary Matrix	**CSV Spreadsheet**	Exports consolidated department statistics: Total students, Placed count, FTE/PPO split, Remaining, and SI offer counts.
Raw JSON Intelligence Dump	**JSON Schema**	Exports full raw structured JSON data for API integrations or external Power BI Desktop data model ingestion.

6. Annual Rollover & Template Ingestion Protocol

To update the portal for subsequent graduating cohorts (e.g., Class of 2027), follow this standardized protocol:

Step 1: Download Standard Template: Open the **'Template Manager'** tab and click *'Download Official CSV Template'* to obtain the verified header structure.

Step 2: Prepare Cohort Master Sheet: Populate columns maintaining standard fields: S.No., Enrollment No., Student Name, Department, Gender, CGPA, Backlogs, Summer Internship Status, Placement Status, Company, and CTC (LPA).

Step 3: Upload via Template Manager: Drag-and-drop the updated CSV file directly into the portal. The built-in parser validates column mappings, computes new aggregates, and refreshes all visualizations immediately.

Step 4: Permanent Codebase Update: Run the Python conversion utility: `python generate_json_v2.py` to recompile `spsu_cdc_data.json` for permanent git commits.

7. Hosting Infrastructure & Offline Execution

The portal is architected with dual-availability infrastructure ensuring 100% operational uptime:

Channel	URL / Location	Operational Notes
Live Web Portal (Cloud)	https://sarlayash.github.io/SPSU-Career-Development-Cell-CDC-Executive-Portal/	Globally accessible 24/7 via secure HTTPS. Powered by GitHub
Offline Standalone File	standalone_dashboard.html (in project directory)	Self-contained 714 KB single-file application. Double-click to run in
Source Code Repository	https://github.com/sarlayash/SPSU-Career-Development-Cell-CDC-Executive-Portal	Private enterprise repository containing source code, data scripts

8. Troubleshooting & Frequently Asked Questions

Q: The browser displays an older version of the portal after deployment.
Fix: Perform a hard cache refresh by pressing **Ctrl + Shift + R** (Windows) or **Cmd + Shift + R** (Mac). The application bundles cache-busting asset hashes.

Q: I forgot or need to change the Administrator credentials.
Fix: Default credentials are **`ADMIN`** / **`SPSU123`**. To update credentials, edit `src/components/LoginPage.jsx` in the source code and rebuild with `npm run build`.

Q: How do I export an executive presentation memo without browser headers and URLs?
Fix: In the print preview window, open 'More Settings' and uncheck 'Headers and footers'. The portal's built-in print CSS automatically inserts institutional headers, banner, and signature blocks.

9. Institutional Authorization & Directorate Sign-Off

This document establishes the official operational standard for the SPSU CDC Executive Intelligence Portal. Any structural modification to database schemas or placement reporting formulas must receive prior approval from the Directorate.

Prepared by:	Reviewed by:	Authorized by:
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